

Manual

Online Visualization of Process Data PDV-OVP 8.2

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1 Online Visualization of Process Data

Overview

Purpose

The PDV-OVR function package includes functions for online visualization of data in the MOC client. Connections can be established to all PDV data servers available in the HYDRA network accessible from the infrastructure. Data from these servers is visualized online. The logical channels and collection requests currently running on the machines specify which data can be displayed.

Existing display instruments and a trend graphic are available to visualize data. Special layouts visualizing facilities or parts thereof can be integrated into the system using suitable editors. Thus creating what is effectively a machine cockpit allowing to view process data of facilities.

Integration

This function package requires a machine interface for data collection in the HYDRA Process Communication Controller (PCC), the integrated data server, the network infrastructure and the basic PDV-PDM package including collection rules.

Features

- Permanent display of process data for one or more machines and facilities. If you have the proper license, you can generate images of facilities on your own (development license required) or you can order HYDRA Customizing and have them generated by MPDV (including runtime license for process visualization).
- Measured values are displayed online with a predefined pointer graphic for a machine. The displayed characteristics can be selected interactively.
- Measured values are displayed online showing a trend line for a machine. You can select the displayed characteristic interactively. The function visualizes the progress and the last measured values for this characteristic online as a continuous trend graphic.

2 Process Trend

Summary

Menu	Quality management → Process data collection→ Process trend
Transaction code	tvpd
Function authorization	tviewpd

The process trend is used to perform process monitoring in quality management.

Usage

The collection processes defined through the collection rules allow to visualize the defined characteristics online. The characteristics identified for visualization are provided to the network by the related data servers. The visualization components existing in this application connect to these servers and will show the measured values online.

The presentation is made as trend line.

Integration

The display functions will visualize the data of the involved shop floor servers in accordance with the collection specifications . In doing so, always a certain amount of historic measured values will be shown and updated by a current value.

Requirements

It must be possible to reach the shop floor servers involved in the network and they must be configured for the presentation of the process data. A registration service (yellow pages to visualize process data online) must be set-up in the network. The display layouts must be available locally or on the server.

Selection parameters

The following selection criteria are available in the application:

Workplace

The pool application workplace can be used to select the desired workplace.

Process parameter

The process parameter drop-down list offers a selection of process parameters.

Workplace

The pool application workplace can be used to select the desired workplace.

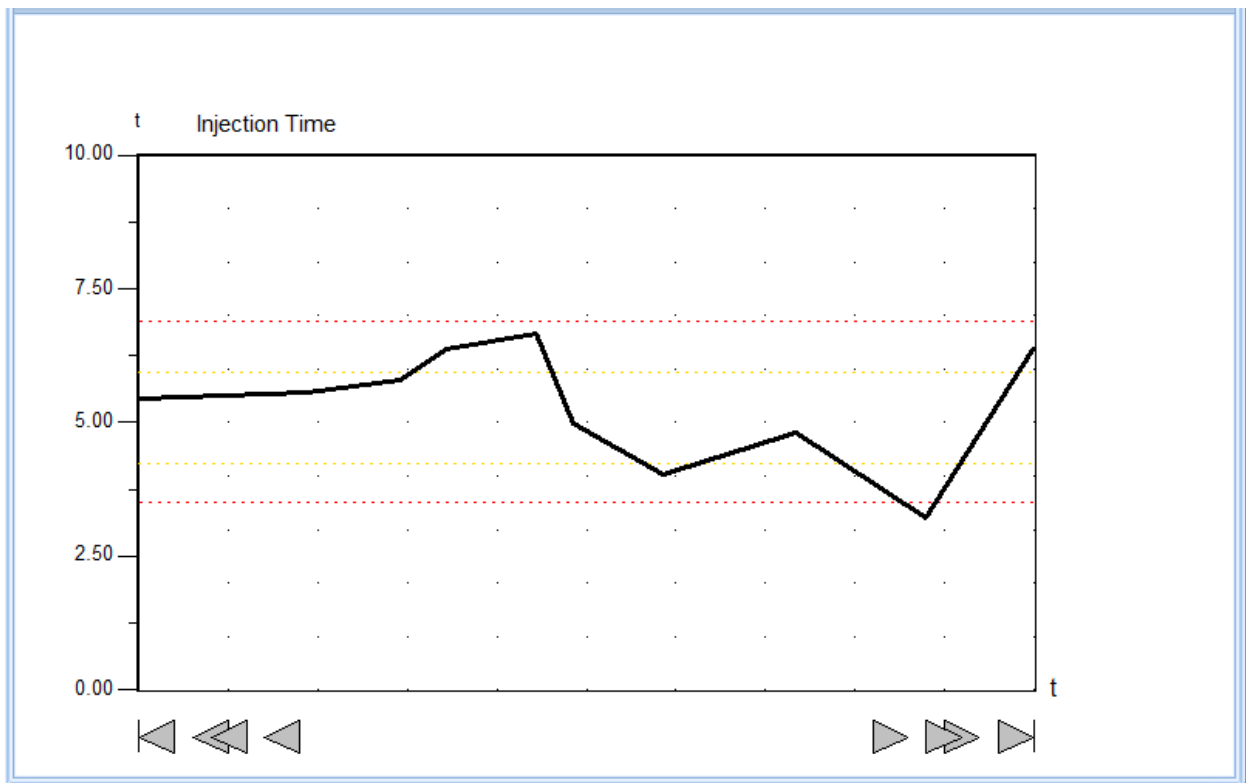
Process parameter

The process parameter drop-down list offers a selection of process parameters.

Field descriptions

Detail applications

The visualization area is used to show the graph of the selected workplace and the individual selected process parameters.



3 Process Data Monitor

Summary

Menu	Quality management → Process monitoring → Process data monitor
Transaction code	mvpd
Function authorization	mviewpd

The process data monitor is used for the process monitoring in quality management.

Usage

The collection procedures defined by the collection rules also allow the online visualization of defined characteristics. The characteristics identified for visualization are provided by the respectively related data servers in the network. The display components existing in this application are connected to these servers and display the measured values online.

The display takes the form of a vector diagram.

Integration

The data of the participating shop floor servers is visualized by the display function based on the specifications of the collection requests. A certain number of measured values from the past are always displayed as well and are part of a continuation including the current value.

Requirement

The participating shop floor servers in the network must be accessible and they must be configured for the presentation of process data. A registration service (yellow pages for process data online visualization) must be set up in the network. The layouts for the display must be available locally or on the server.

Selection criteria

The following selection criteria are available in the application:

Workplace

In the workplace pool application, the desired workplace can be selected.

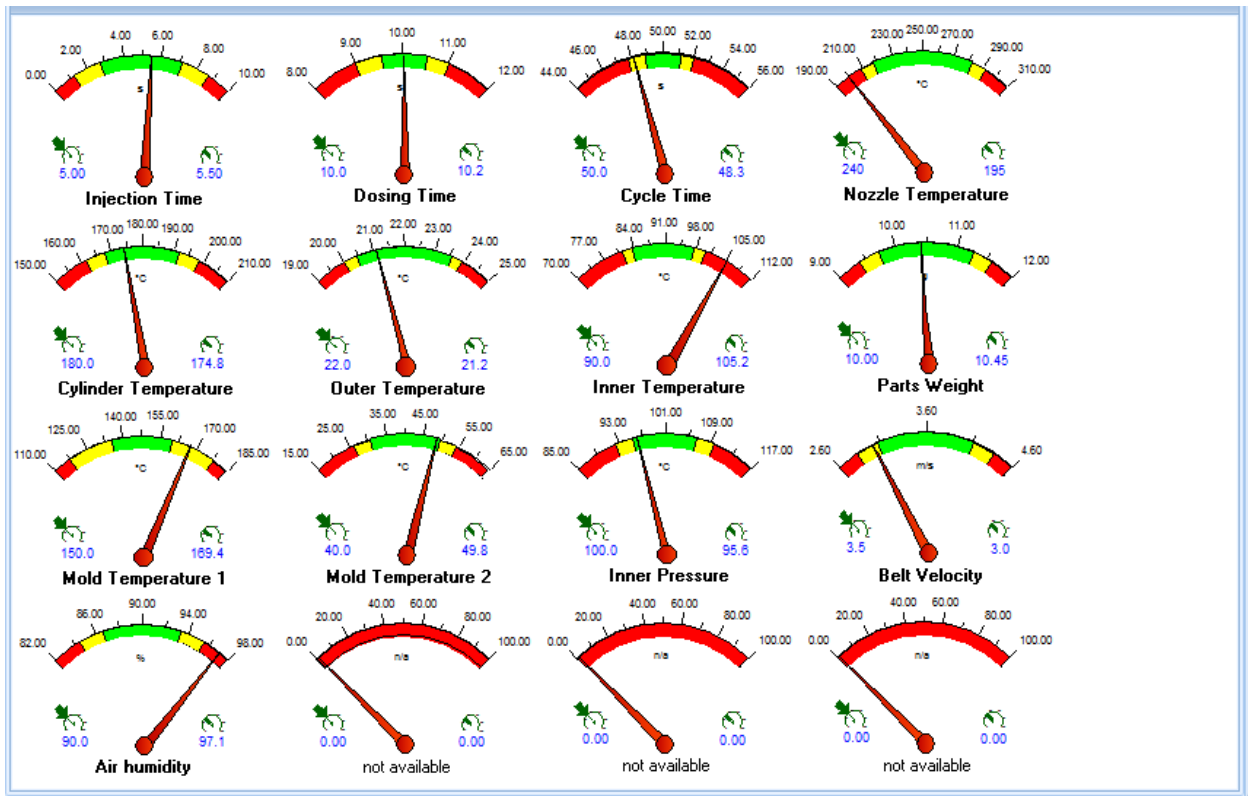
Process parameters

The process parameters drop down list includes a selection of process parameters.

Field descriptions

Detail applications

The selected process parameters pertaining to the selected workplace are displayed in the graphic of the visualization area.



4 Process Visualization

Summary

Menu	Quality management → Process analysis → Process visualization
Transaction code	vispd
Function authorization	visupd

Process visualization is used to monitor the process data in quality management.

Usage

The collection processes defined through the collection rules allow to visualize the defined characteristics online. The related data server provides the characteristics identified for visualization to the network. The visualization components existing in this application connect to these servers and will show the measured values online.

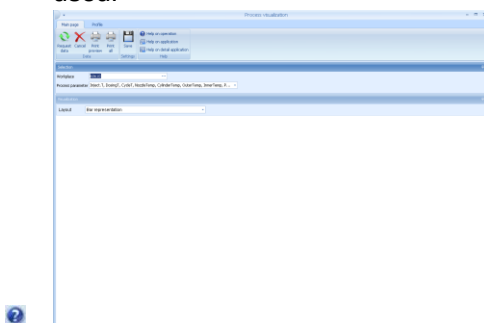
It is also possible to select the display layout and to integrate customized display layouts.

Integration

The display functions will visualize the data of the involved shop floor servers in accordance with the collection specifications.

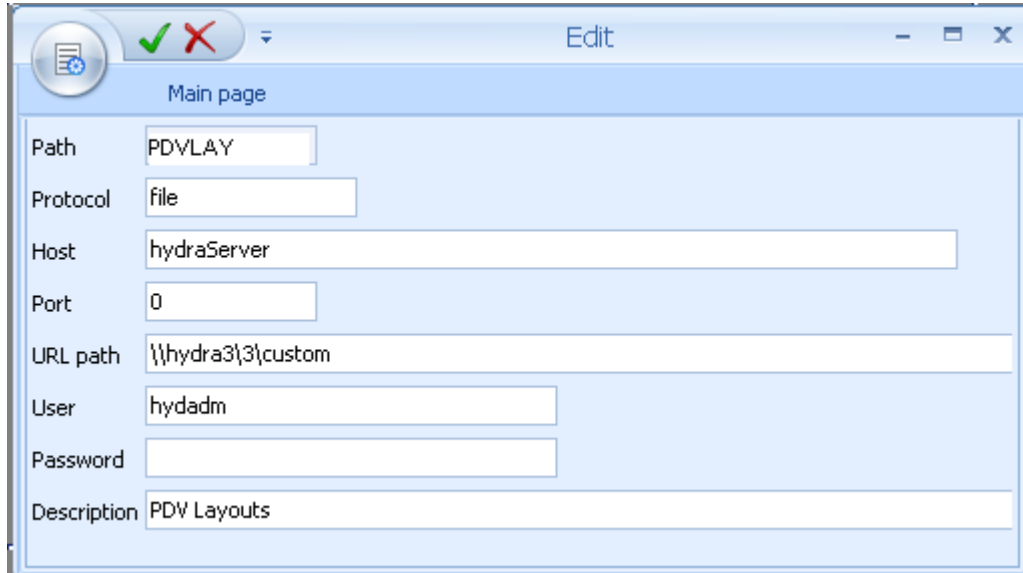
Requirements

- It must be possible to reach the shop floor servers involved in the network and they must be configured for the presentation of the process data. A registration service (yellow pages to visualize process data online) must be set-up in the network. The display layouts must be locally available. If they are not saved locally they will be transferred from the directories of the corresponding HYDRA server to MOC upon the user's request. To do so, the button "Load PDV layouts from server" must be used.



In order to load the data from the server, the HYDRA path "PDVLAY" must be configured. HYDRA paths can be created using the MOC system under "System administration-> System settings -> Paths".

URL path = ./<system no.>/custom/



Selection criteria

The following selection criteria are available in the application:

Workplace

The pool application workplace can be used to select the desired workplace.

Process parameter

The process parameter drop-down list offers a selection of process parameters.

Field descriptions

Detail applications

The visualization area is used to show the graphs of the selected workplace and the individually selected process parameters. The display offers different visualization options that can be selected from the drop-down layout list. Several standard layouts are stored here but also special machine layouts created for the installations and machines.

Examples:

